
## CONTACT

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## TECHNICAL SKILLS

- Python for instrument control and data analysis
- Hardware Design Programming Language
- System Requirement Design
- Mathematical Methods and Data Analysis using Python
- Printed Circuit Board Design with Altium Designer
- Analog Circuit Design (Filters, Operational Amplifiers)
- Scientific Technical Writing
- Matlab Scripting
- Lab test bench setup and configuration
- Hardware/Software Integration for IC testing and validation
- Lab Equipment and Tools: NI TestStand, Digital Pin Map configuration, VSF backend files
- Version Control: GitLab, SVN (Subversion)
- Test Automation
- Development of test sequences in Code Beamer
- PDL (Procedural Description Language) code generation
- MATLAB/Simulink
- Simulation and modeling
- Control system design and analysis
- Flask, FastAPI and SQLite


# ADEYEMI MAYOWA


# SYSTEM TEST ENGINEER

## PROFILE SUMMARY


Test Engineer with an MSc in System Test Engineering (Highest Distinction, FH Joanneum) and hands-on experience spanning semiconductor IC validation and automotive-grade flexible sensor testing. Currently developing test plans, automated test fixtures, and validation infrastructure for flexible sensors at FLEXOO GmbH, working directly with test and requirements engineers at automotive partners including Tesla and Forvia. Delivered a master's thesis at AMS-OSRAM that automated test code generation using PDL, Python, and Jinja cutting test program development time from roughly two weeks to under five seconds. Skilled in test plan development (DV/PV), automated test fixture and script design, DAQ/sensor system integration, root-cause failure analysis, and compliance with automotive quality standards (IATF 16949).


## EDUCATION

**FH Joanneum University of Applied Sciences, Graz, Austria**
*MSc, System Test Engineering* — **Graduated with Highest Distinction**
2023 – 2025 | CGPA: 1.98

- **Master's Thesis**: *Deployment of PDL Flow Within the IC Development: A Unified Approach to Test Code Generation, Validation, and Traceability*
- **Highlight**: Developed Python-Jinja pipeline to automate generation of test skeleton files and component configs, reducing manual effort across 3 chip projects (AS5117, Positron, MOUN)

**COMPUTER ENGINNERING B.Tech**
Ladoke Akintola University Of Technology.
2011 – 2016 (Completed)

## PROFESSIONAL EXPERIENCE

**Test Engineer** *May 2026 – Present*
*FLEXOO GmbH, Heidelberg, Germany*

- **Test Plan Development:** Defining testing parameters and create structured Design Verification / Process Validation (DV/PV) test plans for flexible sensor products, including pressure sensors, temperature sensors, seat sensors, and battery impact detection sensors, working in direct partnership with test and requirements engineers at automotive OEM/Tier-1 partners including Tesla and Forvia.
- **Accelerated Life & Environmental Testing:** Conduct accelerated aging tests using environmental test chambers to validate and verify the long-term functionality and reliability of flexible sensors under extended operating conditions.
- **Test Fixture & Automation Design:** Design custom measurement fixtures for MultiMoS readout hardware for flexible sensors, and develop automated test scripts in Python to streamline sensor characterization and validation.
- **System Integration & Diagnostics:** Integrate flexible sensors with Data Acquisition (DAQ) systems and measurement instrumentation to monitor and analyze key variables including temperature, pressure, and voltage.
- **Failure & Root-Cause Analysis:** Conduct stress and environmental testing, analyze data trends, and pinpoint root causes of component failure to guide corrective actions, including load-cell testing to evaluate pressure response performance of CATL battery cells.
- **Quality & Standards Compliance:** Ensure tested electronic components and sensors comply with industry standards, regulatory requirements, and safety guidelines, working toward automotive test standard IATF 16949.
- **Production Support & Calibration:** Maintain accurate calibration of test equipment and support the transition of sensor products from prototype development to scaled manufacturing.

## CERTIFICATION

Infineon Winter School: *Shaping the Future Through Data Engineering*, 2025

Udemy: *Mastering Selenium: Web Automation Essentials*, 2024

Internet Security ByHassoPlattner Institute, Potsdam, Germany 2018

Hasso Plattner Institute: *Internet Security*, 2018

ACM: *Big Data Fundamentals*, 2017

## SOFT SKILLS

- Strong analytical and problem-solving abilities
- Excellent communication and teamwork skills
- Detail-oriented and proactive approach to tasks
- Analytical Thinking
- Problem-solving in system test automation
- Reporting and documenting test results and system behaviors

## LANGUAGES & INTERESTS

English – (Native), German A1 (Completed)

Interests: IC validation, test automation, test of embedded systems, control theory

## REFRENCES

Available On Request

**Test Data & Software Development Intern** Hycenta GmbH (COMET Center), Graz, Austria | June 2025 – August 2025 *Area 2: Green Energy & Industry – Focus on Hydrogen Storage & Electrochemical Systems*

- **Database & Web Application Development:** Designed and built a secure, full-stack internal web application (Flask, SQLite) from the ground up to centralize hydrogen component data, replacing a company-wide open-access Excel system with **2,000+ records** with an access-controlled, role-based platform used across the testing, research, and customer-facing departments.
- **Access Control & Usability:** Implemented an admin-configurable authentication and permissions system to restrict data access by role, closing a gap where sensitive records had been freely accessible regardless of clearance, and built a simplified, filterable HTML dashboard for fast search, upload, and record management, reducing retrieval time versus manual Excel lookups.
- **Sensor Data Analysis:** Processed and analyzed time-series sensor data from **6 electrolyzer test beds** using Python (Pandas, NumPy) to evaluate performance and identify degradation and correlation patterns for ML readiness.
- **ML Dataset Preparation:** Curated and prepared the full set of company-generated datasets for a **FFG (Austrian Research Promotion Agency)-funded PhD research project**, supporting ML forecasting models for system health and efficiency prediction.
- **Literature Research:** Conducted literature research supporting the same FFG-funded PhD project, presented in Australia, covering hydrogen production cost modeling, market price dynamics, and the produce-vs-store-then-sell decision for optimizing market entry strategy in the EU hydrogen sector.

**AMS-OSRAM** | Graz, Austria | *June 2024 – May 2025*
**Junior Validation & Test Development Engineer** *(progressed from Test & Validation Engineer Intern)*
**Test & Validation Engineer Intern** | *June 2024 – August 2024*

- **Python OOP & Instrument Control:** Developed Python OOP scripts for automated SCPI-based instrument control, improving measurement efficiency and repeatability across lab test benches.
- **PDL Test Sequencing & Version Control:** Implemented Python test sequences from PDL/Code Beamer and managed script baselines using SVN and GitLab, establishing a stable, traceable code environment for validation engineers.
- **Lab Bench Support & Documentation:** Supported lab bench setup (register/pin maps, relay matrix, NI tools) and contributed to test specification documents and engineering support for lab engineers.

**Junior Validation & Test Development Engineer (Master's Thesis)** — *Dec 2024 – May 2025*

- **Test Program Generation Speed-up:** Developed automated test flow sequences from PDL to Python/C++ for IC validation on Diamondx ATE testers reducing test program generation in UNISON C from a **~2-week manual development cycle to under 5 seconds**, via the automated PDL→Python/Jinja pipeline.
- **Test Data, Software & ATE Integration:** Designed and deployed automated test flow sequences (PDL → Python → C++) on Diamondx ATE and Linux simulator environments, enabling early, efficient IC validation across **3 chip projects (AS5117, Positron, MOUN)**.
- **Framework Optimization:** Integrated and optimized automated test frameworks, improving traceability and repeatability while resolving key bottlenecks in the validation workflow.
- **Lab Infrastructure & Collaboration:** Enhanced lab validation infrastructure by generating Python test sequences, managing version control (SVN/GitLab), and collaborating with lab engineers on bench configuration and documentation.